

Harsh Vishwakarma

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GitHub | LinkedIn | LeetCode | Portfolio

PROFILE

Backend engineer and founding member of a cloud-management startup. Specializes in AWS automation and distributed systems, building high-performance SaaS platforms using Python, FastAPI, and asynchronous microservices. Experienced in cloud governance, cost-optimization, and multi-region resource visibility across 250+ AWS services.

EDUCATION

Graphic Era University

B.Tech in Computer Science | CGPA: 8.09/10.00

2021 – 2025

Dehradun

Woodland Academy, C.B.S.E

12th Grade | 86.8%

2020 – 2021

Gorakhpur

PROFESSIONAL EXPERIENCE

CloudLaminate, BlueConcur Private Limited

Software Engineer I

May 2025 – Present

Remote

AWS Native Services Platform

- Launched an async microservice backend (FastAPI, aioboto3) for a multi-tenant cloud governance platform covering **250+ AWS services**, cutting API latency by **over 60%**.
- Developed a native services engine with metadata enrichment and FinOps integration, attaching per-resource pricing and hardware metrics to improve cost-transparency accuracy by **~40%**.
- Implemented bulk export pipelines (CSV/JSON) for organization-wide resource inventories, reducing console-navigation overhead by **70%+** and speeding up audit workflows by **3x**.

AWS Cost Calculator

- Devised a configuration-normalization engine that expands workload specs into infrastructure blueprints for EC2, S3, and EKS, eliminating dozens of manual configuration steps.
- Implemented a deterministic transformation pipeline to generate schema-compliant specs for AWS Price List API ingestion, improving cross-team cost-forecasting reliability by **30–50%**.
- Enabled enterprise what-if cost simulations for CI/CD and IaC pipelines, cutting infrastructure estimation time for DevOps and Finance teams by **80%**.

PROJECTS & TECHNICAL EXPERIENCE

Distributed Multi-Drone AI Tracking System

2024 – 2025

- Designed a distributed edge-computing setup using Python and MAVSDK, offloading YOLOv8 inference to an Nvidia Jetson while a Raspberry Pi handled real-time autonomous flight control.
- Constructed a multi-stream computer vision pipeline with GStreamer and Nvidia DeepStream to simultaneously decode and analyze multiple 1280×960 RTSP drone feeds with NvDCF pixel-tracking.
- Engineered a 10Hz UDP communication bridge transmitting PID-calculated velocity vectors via MAVLink, with asynchronous state machines and timeout-to-hover failsafes on network loss.

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript, SQL, HTML/CSS
- Backend & APIs:** FastAPI, REST API design, Boto3/aioboto3, UDP/Socket Programming, WebSocket
- Cloud & DevOps:** AWS (EC2, S3, EKS, Price List API), Git, Linux
- Computer Vision & AI:** YOLOv8, GStreamer, Nvidia DeepStream, NvDCF, MAVLink, MAVSDK
- Databases:** MySQL, PostgreSQL
- Frontend:** Angular, HTML/CSS, JavaScript
- Core CS:** Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS

PUBLICATIONS

Assessing the Accuracy of Regression and Classification Models in Air Quality Prediction, IEEE

2024 – 2025

Proposed an ML-based pollution forecasting model improving prediction accuracy by **20%** over traditional baselines.

Sentimental Analysis of Movie Reviews using Naive Bayes and Random Forest, IEEE

2022 – 2023

Achieved **95%+** accuracy in sentiment classification, surpassing prior benchmarks by **15%**.

ACHIEVEMENTS & EXTRA-CURRICULAR

G20 Symposium on Digital Agriculture (2023): Finalist (Runner-up) at Graphic Era University, competing on IoT applications in agriculture as part of Team Triad.

2023

NSS Certificates: Earned NSS 'B' and 'C' Certificates through community projects and leadership roles.

2022 – 2024

Competitive Programming: Solved **500+ problems** on LeetCode, GFG, and HackerRank with focus on DSA.

Ongoing